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FORMING DEVICE AND FORMING METHOD THEREOF

Abstract

A forming device includes a feeding component, an injection module, and a die-casting module. The feeding component has an internal space suitable for allowing a molten metal fluid to flow. The injection module is disposed on the feeding component and communicated with the internal space. The die-casting module has a mold cavity and a flow channel. The flow channel is aligned with the mold cavity and communicated with the internal space of the feeding component. The molten metal fluid flows to the die-casting module along the internal space, the injection module outputs a supercritical fluid into the internal space to mix with the molten metal fluid, and the feeding component pushes the molten metal fluid to pass through the flow channel and enter the mold cavity.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113107687, filed on Mar. 4, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a die-casting molding process, and particularly relates to a forming device and a forming method thereof suitable for metal alloys.

Description of Related Art

[0003] With the demand for portability, the demand for thinner and lighter laptops is also increasing. Currently, the mainstream appearance material technology for slim laptops is primarily based on computer numerical control (CNC) processing of aluminum alloy stamping or aluminum extrusion. However, the steps involved in CNC processing are complicated and costly. Adopting the die-casting molding process can improve the CNC processing steps and reduce costs.

[0004] Existing magnesium alloy die-casting parts have low density, providing an advantage in lightweight materials. However, achieving wall thicknesses below 0.8 mm is limited by process constraints, usually resulting in structural shrinkage or post-shrinkage crack defect phenomena, which makes it impossible to achieve a design with a thickness comparable to aluminum alloy, making it challenging for magnesium alloy materials to be fully utilized in thin laptops.

[0005] The magnesium alloy casing of existing thin laptops can be produced using semi-solid metal die casting, while casings without complicated appearance or non-thin appearance parts can be made using the cold-chamber die-casting process. In addition, the hot-chamber die-casting process is applied to non-appearance parts for thin laptops. Nowadays, the requirements for thinness in various types of casings, parts, or appearance parts have increased significantly. Since the thinning of workpieces is closely related to the flow rate of metal fluid, how to improve the flow rate in metal fluid in semi-solid die casting, cold-chamber die casting, and hot-chamber die casting has become an important development goal.

SUMMARY

[0006] The disclosure provides a forming device and a forming method thereof. During the die-casting process, a supercritical fluid is injected to mix with a molten metal fluid, thereby increasing the flow rate of the molten metal fluid to achieve the purpose of thin die casting.

[0007] The forming device according to the disclosure includes a feeding component, an injection module, and a die-casting module. The feeding component has an internal space suitable for allowing a molten metal fluid to flow. The injection module is disposed on the feeding component and communicated with the internal space. The die-casting module has a mold cavity and a flow channel. The flow channel is aligned with the mold cavity and communicated with the internal space of the feeding component. The injection module outputs a supercritical fluid into the internal space to mix the molten metal fluid, and the feeding component pushes the mixture of the molten metal fluid and the supercritical fluid to pass through the flow channel and enter the mold cavity.

[0008] The molding method according to the disclosure is suitable for a forming device, and the forming device includes a feeding component, an injection module, and a die-casting module. The molding method includes: providing a molten metal fluid to the feeding component; flowing, by the molten metal fluid, toward the die-casting module; outputting, by the injection module, a supercritical fluid to the feeding component to mix with the molten metal fluid; pushing, by the

feeding component, the molten metal fluid to pass through a flow channel and enter a mold cavity of the die-casting module; and pressurizing, by the die-casting module, the molten metal fluid to form a solid workpiece.

[0009] Based on the above, the forming device and the forming method thereof according to the disclosure utilize the supercritical fluid injected into the feeding component, allowing to mix with the molten metal fluid. Through utilizing the characteristics of the supercritical fluid such as gas-like viscosity and liquid-like density, which results in low surface tension and high transmission efficiency, also the solubility of the supercritical fluid can be adjusted with temperature and pressure, thereby the flow rate of the molten metal fluid during the die-casting process can be enhanced, the flow length of the molten metal fluid can be increased, and the pressure required for die-casting can be reduced. In addition, the supercritical fluid in the mold cavity changes into a gaseous state due to the temperature and pressure drop, which reduces the internal pressure of the mold cavity, thereby improving the shrinkage problem of the metal alloy after die-casting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1A is a schematic structural diagram of a forming device according to the first embodiment of the disclosure.

[0011] FIG. 1B is a flow block diagram of a forming method of the disclosure.

[0012] FIG. 1C is a flow block diagram of another forming method of the disclosure.

[0013] FIG. 2A is a schematic structural diagram of a forming device according to the second embodiment of the disclosure.

[0014] FIG. 2B is a schematic diagram of a feeding action of the forming device in FIG. 2A.

[0015] FIG. 3A is a schematic structural diagram of a forming device according to the third embodiment of the disclosure.

[0016] FIG. 3B is a schematic diagram of a feeding action of the forming device in FIG. 3A.

DESCRIPTION OF THE EMBODIMENTS

[0017] FIG. 1A is a schematic structural diagram of a forming device according to the first embodiment of the disclosure. FIG. 1B is a flow block diagram of a forming method of the disclosure.

[0018] Referring to FIG. 1A, a forming device **100** of the disclosure is suitable for die-casting of metal alloys, and the forming device **100** includes a feeding component **110**, an injection module **120**, and a die-casting module **130**. The feeding component **110** has an internal space IS for allowing a molten metal fluid MF to flow. The injection module **120** is disposed on the feeding component **110** and communicated with the internal space IS. The die-casting module **130** has a mold cavity **131** and a flow channel **132**. The mold cavity **131** is used to die-cast a workpiece shape according to needs, and the flow channel **132** is aligned with the mold cavity **131** and communicated with the internal space IS of the feeding component **110**.

[0019] First, a molten metal fluid MF is driven by external force and flows along the internal space IS to the die-casting module **130**, then the injection module **120** outputs a supercritical fluid SF during the flow process into the internal space IS to mix with the flowing molten metal fluid MF, and the feeding component **110** pushes the molten metal fluid MF to pass through the flow channel **132** and enter the mold cavity **131**.

[0020] Furthermore, the molten metal fluid MF adopts magnesium alloy, aluminum alloy, or other similar alloy materials. Moreover, the molten metal fluid MF is a solid metal before heating, and the solid metal forms a molten state after heating and is introduced into the feeding component **110** and then pushed to the mold cavity **131** of the die-casting module **130** by applying external force.

[0021] In the embodiment, the supercritical fluid SF adopts nitrogen, carbon dioxide, or a mixture

of multiple gases. The characteristic of the supercritical fluid SF is that the properties of gaseous and liquid states are closely similar, resulting in a fluid phenomenon of uniform phase. Therefore, the supercritical fluid SF has the compressibility of a gas, but also has fluidity close to a liquid, and the density of the supercritical fluid SF is of 0.1 to 1.0 g/ml.

[0022] Referring to FIG. 1B, a specific forming method of the forming device **100** of the disclosure is described below. Step S1: A molten metal fluid MF is provided to the internal space IS of the feeding component **110**. Step S2: The molten metal fluid MF is driven by external force and flows toward the die-casting module **130**. Step S3: The supercritical fluid SF is output by the injection module **120** to the feeding component **110** to mix with the molten metal fluid MF. Step S4: A mixture of the molten metal fluid MF and the supercritical fluid SF is pushed by the feeding component **110** to pass through the flow channel **132** and enter the mold cavity **131** of the die-casting module **130**. Step S5: The mixture of the molten metal fluid MF and the supercritical fluid SF is pressurized by the die-casting module **130** to form a solid workpiece. Finally, after the pressure and temperature of the molten metal fluid MF decrease, the solid workpiece may be removed, and the aforementioned forming steps may be repeated.

[0023] In addition, the forming method is suitable for the thick-wall molding process (greater than 3 mm). Since the thick-wall finished product requires a large tonnage of external force for one-piece die casting and needs to be pushed over a longer distance, this embodiment utilizes the supercritical fluid assists to assist in pushing the molten metal fluid. Additionally, the mixture of the supercritical fluid and the molten metal fluid can be pushed into the mold cavity together by the feeding component, resulting in a product filled with bubbles (as the supercritical fluid transitions to a gaseous state due to temperature and pressure drops in the mold cavity), thereby reducing the required material weight and material cost, and reducing processing time and energy consumption.

[0024] Furthermore, the forming device **100** of the disclosure is also suitable for the thin-wall molding process (less than 3 mm).

[0025] Referring to FIG. 1C, a specific forming method of the forming device **100** of the disclosure being applied to the thin-section forming process (less than 3 mm) is described below. Step T1: The molten metal fluid MF is provided to the internal space IS of the feeding component **110**. Step T2: The molten metal fluid MF is driven by external force and flows toward the die-casting module **130**. Step T3: After a portion of the molten metal fluid MF fills 60%-80% of the mold cavity **131**, the injection module **120** outputs the supercritical fluid SF to the feeding component **110** to mix with the molten metal fluid MF. Step T4: The supercritical fluid SF assists in pushing another portion of the molten metal fluid MF to pass through the flow channel **132** to fill the mold cavity **131**. Step T5: The molten metal fluid MF is pressurized by the die-casting module **130** to form a solid workpiece.

[0026] In short, during the thin-wall molding process, the supercritical fluid does not enter the mold cavity. Only after a portion of the molten metal fluid enters the mold cavity, then the injection module injects the supercritical fluid, which then pushes another portion of the molten metal fluid located in the flow channel **132** to fill the mold cavity, thereby increasing the flow rate of the molten metal fluid.

[0027] Referring to FIG. 1A, this embodiment shows a forming device **100** using the supercritical fluid to assist in metal semi-solid forming, in which the feeding component **110** includes a delivery pipe **111**, a feeding member **112**, and a pushing member **113**. The feeding member **112** receives a plurality of metal particles MP, the delivery pipe **111** has the internal space IS, and the feeding member **112** is disposed outside the delivery pipe **111** and communicated with the internal space IS. Specifically, the feeding member **112** adopts a funnel, and the pushing member **113** is movably disposed in the delivery pipe **111** and is located below the feeding member **112**.

[0028] The forming device **100** includes a plurality of heating members **140** disposed on an outer wall surface of the delivery pipe **111** for heating the plurality of metal particles MP and the molten metal fluid MF.

[0029] Referring to FIG. 1A, the internal space IS is divided into a storage area P1, a transition area P2, and a mixing area P3. The multiple metal particles MP pass through the feeding member 112 and enter the storage area P1 of the internal space IS. The storage area P1 is used to store the multiple metal particles MP, and during the movement process of the metal particles MP, the metal particles MP are gradually heated by the multiple heating members 140. In short, the storage area P1 is gradually heated by the heating member 140. The transition area P2 is adjacent to the material storage area P1 and the multiple metal particles MP form the molten metal fluid MF after entering the transition area P2. The mixing area P3 is adjacent to the transition area P2 and close to the die-casting module 130. When the molten metal fluid MF flows from the transition area P2 and enters the mixing area P3, the injection module 120 outputs the supercritical fluid SF to flow into the mixing area P3, allowing to mix with the molten metal fluid MF, and the supercritical fluid SF can also assist in pushing the molten metal fluid MF located in the mixing area P3.

[0030] In addition, the injection module 120 is disposed with a check valve to prevent the supercritical fluid SF from flowing back and causing a local pressure drop in the mixing area P3.

[0031] Referring to FIG. 1A, the pushing member 113 is a hollow screw rotatably disposed in the storage area P1 of the internal space IS, and the hollow screw is suitable for rotating to drive the multiple metal particles MP to move toward the transition area P2.

[0032] In detail, the hollow screw exerts external force during the rotation process, thereby pushing the mixture of the molten metal fluid MF and the supercritical fluid SF from the mixing area P3 into the mold cavity 131. Since the mixture of the molten metal fluid MF and the supercritical fluid SF has characteristics such as low viscosity, low surface tension, and high transmission efficiency, the flow length of the molten metal fluid MF can be enhanced, and the die-casting time of the molten metal fluid MF can be reduced. In addition, the supercritical fluid SF changes into a gaseous state due to the temperature and pressure drop in the mold cavity 131, which can reduce the internal die-casting pressure of the mold cavity 131, thereby improving the volume shrinkage problem of the metal material during die-casting.

[0033] Referring to FIG. 2A and FIG. 2B, this embodiment shows a forming device 100A using the supercritical fluid to assist in metal cold-chamber die casting, in which a feeding component 110a includes a delivery pipe 111a, a feeding member 112a, and a pushing member 113a. The feeding member 112a receives the molten metal fluid MF, the delivery pipe 111a has the internal space IS, the feeding member 112a is disposed outside the delivery pipe 111a and communicated with the internal space IS. Specifically, the feeding member 112a adopts a funnel, and the pushing member 113a is movably disposed in the internal space IS and is located below the feeding member 112a.

[0034] The forming device 100A includes a plurality of heating members 140a disposed on the outer wall surface of the delivery pipe 111a for heating the molten metal fluid MF to maintain the fluidity thereof.

[0035] Referring to FIG. 2A, the internal space IS is divided into the storage area P1, the transition area P2, and the mixing area P3. The molten metal fluid MF enters the storage area P1 of the internal space IS from the feeding member 112a. The storage area P1 is used to store the molten metal fluid MF, and during the movement process, the molten metal fluid MF is continuously heated by the multiple heating members 140a, so as to maintain the high temperature and fluidity of the molten metal fluid MF. The transition area P2 is adjacent to the storage area P1 and is disposed with a check valve VL. The transition area P2 is suitable for allowing the molten metal fluid MF to pass through. The mixing area P3 is adjacent to the check valve VL and close to the die-casting module 130. When the molten metal fluid MF flows from the storage area P1, passes through the check valve VL of the transition area P2, and enters the mixing area P3, the injection module 120a outputs the supercritical fluid SF to flow into the mixing area P3, allowing to mix with the molten metal fluid MF, and the supercritical fluid SF can also assist in pushing the molten metal fluid MF located in the mixing area P3.

[0036] In addition, the injection module 120a is disposed with a check valve to prevent the

supercritical fluid SF from flowing back and causing a local pressure drop in the mixing area P3. [0037] The pushing member **113a** is a pushrod linearly movably disposed in the storage area P1 of the internal space IS and away from the die-casting module **130a**. The pushrod is suitable for pushing the molten metal fluid MF to pass through the check valve VL and enter the mixing area P3.

[0038] Referring to FIG. 2B, in detail, when moving toward the check valve VL, the pushrod exerts external force, thereby pushing the mixture of the molten metal fluid MF and the supercritical fluid SF from the mixing area P3 into the mold cavity **131**. Since the mixture of the molten metal fluid MF and the supercritical fluid SF possesses characteristics such as low viscosity, low surface tension, and high transmission efficiency, the flow length of the molten metal fluid MF can be enhanced, and the die-casting time of the molten metal fluid MF can be reduced. In addition, the supercritical fluid SF changes into a gaseous state due to the temperature and pressure drop in the mold cavity **131**, which can reduce the internal die-casting pressure of the mold cavity **131**, thereby improving the volume shrinkage problem of the metal material during die-casting.

[0039] Referring to FIG. 3A and FIG. 3B, this embodiment shows a forming device **100B** using the supercritical fluid to assist in hot-chamber die-casting forming, in which a feeding component **110b** includes a delivery pipe **111b**, a feeding member **112b**, and a pushing member **113b**. The delivery pipe **111b** has the internal space IS, and the feeding member **112b** is disposed outside the delivery pipe **111b** and communicated with the internal space IS. The pushing member **113b** is movably disposed on the internal space IS and partially protrudes from the feeding member **112b**.

[0040] The feeding component **110b** includes a heating furnace **150b** for heating and containing the molten metal fluid MF, and the end of the delivery pipe **111b** away from the die-casting module **130b** and the feeding member **112b** are located in the heating furnace **150b**. In this embodiment, the heating furnace **150b** is used to facilitate the replenishment of the molten metal fluid MF. The metal particles MP only need to be placed in the heating furnace **150b** to undergo the heating and melting process, which can simplify the replenishment process of the molten metal fluid MF and is also beneficial to recycling and reusing the metal particle MP.

[0041] The internal space IS of the delivery pipe **111b** is divided into the storage area P1, the transition area P2, and the mixing area P3. The feeding member **112b** is communicated with the heating furnace **150b** and is disposed with a first check valve VL1. The molten metal fluid MF is suitable for flowing from the heating furnace **150b**, passing through the first check valve VL1, and entering the storage area P1. The transition area P2 is disposed with a second check valve VL2 and is suitable for allowing the molten metal fluid MF to pass through and enter the mixing area P3. The mixing area P3 is adjacent to the second check valve VL2 and close to the die-casting module **130b**. When the molten metal fluid MF flows from the storage area P1, passes through the second check valve VL2 of the transition area P2, and enters the mixing area P3, the injection module **120b** outputs the supercritical fluid SF to flow into the mixing area P3, allowing to mix with the molten metal fluid MF, and the supercritical fluid SF can also assist in pushing the molten metal fluid MF located in the mixing area P3.

[0042] Furthermore, the first check valve VL1 can prevent the molten metal fluid MF from flowing back to the heating furnace **150b**, the second check valve VL2 can prevent the molten metal fluid MF and the supercritical fluid SF from flowing back to the storage area P1, and the injection module **120b** is disposed with a check valve to prevent the supercritical fluid SF from flowing back and causing a local pressure drop in the mixing area P3.

[0043] The pushing member **113b** is a pushrod linearly movably disposed on the feeding member **112b** and away from the die-casting module **130b**, and the pushing member **113b** is suitable for pushing the molten metal fluid MF to pass through the second check valve VL2 and enter the mixing area P3. In addition, a moving direction D1 of the pushing member **113b** in the feeding member **112b** is orthogonal to an extending direction D2 of the mixing area P3, referring to FIG. 3B, when the pushing member **113b** moves downward along the movement direction D1, the

molten metal fluid MF is pushed to rise obliquely along the delivery pipe **111b** to pass through the second check valve VL2.

[0044] Referring to FIG. 3B, in detail, when moving downward and passing through the first check valve VL1, the pushing member **113b** exerts external force, thereby pushing the mixture of the molten metal fluid MF and the supercritical fluid SF from the mixing area P3 into the mold cavity **131b**. Since the mixture of the molten metal fluid MF and the supercritical fluid SF possesses characteristics such as low viscosity, low surface tension, and high transmission efficiency, the flow length of the molten metal fluid MF can be enhanced, and the die-casting time of the molten metal fluid MF can be reduced. In addition, the supercritical fluid SF changes into a gaseous state due to the temperature and pressure drop in the mold cavity **131**, which can reduce the internal die-casting pressure of the mold cavity **131**, thereby improving the volume shrinkage problem of the metal material during die-casting.

[0045] In summary, the forming device and the forming method thereof according to the disclosure utilize the supercritical fluid injected into the feeding component, allowing to mix with the molten metal fluid. Through utilizing the characteristics of the supercritical fluid such as gas-like viscosity and liquid-like density, which results in low surface tension and high transmission efficiency, also the solubility of the supercritical fluid can be adjusted with temperature and pressure, thereby the flow rate of the molten metal fluid during the die-casting process can be enhanced, the flow length of the molten metal fluid can be increased, and the pressure required for die-casting can be reduced. In addition, the supercritical fluid in the mold cavity changes into a gaseous state due to the temperature and pressure drop, which reduces the internal pressure of the mold cavity, thereby improving the shrinkage problem of the metal alloy after die-casting. In addition, the forming device and the forming method thereof according to the disclosure can also utilize the supercritical fluid injected into the feeding component, allowing to mix with the molten metal fluid and be injected into the mold cavity together, so that the product is filled with bubbles, thereby reducing the required material weight and material costs, and reducing processing time and energy consumption.

Claims

1. A forming device, comprising: a feeding component having an internal space suitable for allowing a molten metal fluid to flow; an injection module disposed on the feeding component and communicated with the internal space; and a die-casting module having a mold cavity and a flow channel, wherein the flow channel is aligned with the mold cavity and communicated with the internal space of the feeding component, wherein the injection module outputs a supercritical fluid into the internal space to mix with the molten metal fluid, and the feeding component pushes the molten metal fluid to pass through the flow channel and enter the mold cavity.
2. The forming device as claimed in claim 1, wherein the feeding component comprises a delivery pipe, a feeding member, and a pushing member, the delivery pipe has the internal space, the feeding member is disposed outside the delivery pipe and communicated with the internal space, and the pushing member is movably disposed in the delivery pipe.
3. The forming device as claimed in claim 2, wherein the feeding member receives a plurality of metal particles, the internal space is divided into a storage area, a transition area, and a mixing area, the storage area stores the metal particles and the metal particles are gradually heated, the transition area is adjacent to the storage area and the metal particles form the molten metal fluid after entering the transition area, the mixing area is adjacent to the transition area and close to the die-casting module, and the supercritical fluid flows into the mixing area and assists in pushing the molten metal fluid located in the mixing area.
4. The forming device as claimed in claim 3, wherein the pushing member is a hollow screw rotatably disposed in the storage area of the internal space, and the hollow screw is suitable for

rotating to drive the metal particles to move toward the transition area.

5. The forming device as claimed in claim 2, wherein the pushing member pushes a mixture of the molten metal fluid and the supercritical fluid into the mold cavity.

6. The forming device as claimed in claim 2, wherein the feeding member receives the molten metal fluid, the internal space is divided into a storage area, a transition area, and a mixing area, the storage area stores the molten metal fluid, the transition area is disposed with a check valve and is suitable for allowing the molten metal fluid to pass through, the mixing area is adjacent to the check valve and close to the die-casting module, and the supercritical fluid flows into the mixing area and assists in pushing the molten metal fluid located in the mixing area.

7. The forming device as claimed in claim 6, wherein the pushing member is a pushrod linearly movably disposed in the storage area of the internal space and away from the die-casting module, and the pushrod is suitable for pushing the molten metal fluid to pass through the check valve and enter the mixing area.

8. The forming device as claimed in claim 2, further comprising a heating furnace for containing the molten metal fluid, and an end of the delivery pipe away from the die-casting module and the feeding member are located in the heating furnace.

9. The forming device as claimed in claim 8, wherein the internal space is divided into a storage area, a transition area, and a mixing area, the feeding member is disposed with a first check valve, the molten metal fluid flows from the heating furnace, passes through the first check valve, and enters the storage area, the transition area is disposed with a second check valve and is suitable for allowing the molten metal fluid to pass through, the mixing area is adjacent to the second check valve and close to the die-casting module, and the supercritical fluid flows into the mixing area and assists in pushing the molten metal fluid located in the mixing area.

10. The forming device as claimed in claim 9, wherein the pushing member is a pushrod linearly movably disposed on the feeding member and away from the die-casting module, and the pushrod is suitable for pushing the molten metal fluid to pass through the second check valve and enter the mixing area.

11. The forming device as claimed in claim 3, further comprising a plurality of heating members disposed on an outer wall surface of the delivery pipe for heating the metal particles or the molten metal fluid.

12. The forming device as claimed in claim 6, further comprising a plurality of heating members disposed on an outer wall surface of the delivery pipe for heating metal particles or the molten metal fluid.

13. A forming method suitable for a forming device, wherein the forming device comprises a feeding component, an injection module, and a die-casting module, and the forming method comprises: providing a molten metal fluid to the feeding component, wherein the molten metal fluid flows toward the die-casting module; outputting, by the injection module, a supercritical fluid to the feeding component to mix with the molten metal fluid; pushing, by the feeding component, the molten metal fluid to pass through the flow channel and enter a mold cavity of the die-casting module; and pressurizing, by the die-casting module, the molten metal fluid to form a solid workpiece.

14. The molding method as claimed in claim 13, wherein the supercritical fluid adopts nitrogen, carbon dioxide, or a mixture of a plurality of gases.

15. The molding method as claimed in claim 13, wherein the molten metal fluid adopts magnesium alloy or aluminum alloy.

16. The molding method as claimed in claim 13, wherein the feeding component pushes a mixture of the molten metal fluid and the supercritical fluid into the mold cavity.

17. The molding method as claimed in claim 13, wherein the feeding component pushes a portion of the molten metal fluid to partially fill the mold cavity, and the supercritical fluid assists in pushing another portion of the molten metal fluid to fill the mold cavity.

